

Evolution, Paleobotany & Economic Botany

ACADEMIC STUDY NOTES

BOTANY · PALEOBOTANY · ETHNOBOTANY

This comprehensive study guide covers two major thematic units in plant sciences: **Evolution & Paleobotany** and **Economic Botany & Ethnobotany** (with emphasis on Assam). The first unit explores the origin of life, evolutionary theories, natural selection, speciation, fossil evidence, the geological time scale, and the remarkable journey of plant evolution from ancient oceans to modern forests. The second unit examines economically important plant groups — cereals, pulses, fibers, oils, spices, beverages, medicinal plants, and forest products — alongside the rich ethnobotanical traditions of Assam, including its iconic tea and bamboo heritage. These notes are designed for undergraduate and graduate students in botany, paleobotany, and ethnobotany, as well as researchers and educators seeking structured, detailed reference material.

Origin of Life & Theories of Evolution

EVOLUTION & PALEOBOTANY

Origin of Life

The origin of life on Earth remains one of the most profound questions in biology. The earliest evidence of life dates to approximately **3.5–3.8 billion years ago**, based on stromatolites and microfossils found in ancient rock formations. Life is believed to have originated in the primordial oceans through a process called **chemical evolution** or **abiogenesis** — the gradual formation of complex organic molecules from simpler inorganic compounds under the conditions of the early Earth.

The **Oparin-Haldane Hypothesis** (1920s) proposed that the early Earth's atmosphere was reducing, containing methane (CH₄), ammonia (NH₃), hydrogen (H₂), and water vapor (H₂O), but no free oxygen. Energy from lightning, UV radiation, and volcanic activity drove the synthesis of organic compounds, forming a "primordial soup." The famous **Miller-Urey Experiment** (1953) simulated these conditions and successfully produced amino acids, lending experimental support to this hypothesis. Over millions of years, these organic molecules organized into coacervates and eventually into the first primitive cells — **prokaryotes** — capable of metabolism and reproduction.

Spontaneous Generation (Abiogenesis)

Ancient belief that life arises directly from non-living matter. Disproved by Louis Pasteur's swan-neck flask experiment (1861), which demonstrated that microorganisms come from pre-existing microorganisms.

Panspermia Theory

Proposed by Svante Arrhenius (1903), this theory suggests that life originated elsewhere in the universe and reached Earth via meteorites or cosmic dust. While intriguing, it does not explain the ultimate origin of life — only its transfer.

Special Creation

Religious and philosophical view that life was created by a supernatural being. Not a scientific theory as it cannot be tested or falsified, but historically significant in the development of biological thought.

Modern Synthetic Theory

Integrates Darwinian natural selection with Mendelian genetics and molecular biology. Explains evolution as changes in allele frequencies within populations over time, driven by mutation, recombination, gene flow, and natural selection.

Theories of Evolution

Evolution refers to the gradual change in the heritable characteristics of biological populations over successive generations. Several key theories have shaped our understanding of this process. **Lamarck's Theory of Inheritance of Acquired Characteristics** (1809) proposed that organisms pass on traits acquired during their lifetime to offspring — for example, a giraffe stretching its neck to reach leaves would produce offspring with longer necks. While largely discredited, Lamarck was the first to propose a systematic mechanism for evolutionary change.

Darwin's Theory of Descent with Modification (1859) revolutionized biology by proposing that all species share common ancestors and that natural selection drives adaptation. Darwin's observations during the voyage of HMS Beagle, particularly in the Galápagos Islands, provided critical evidence. **Alfred Russel Wallace** independently conceived the same idea, prompting the joint presentation of their theories in 1858. The **Modern Synthesis** (1930s–1950s) unified Darwinian selection with Mendelian genetics, population genetics, and paleontology into a comprehensive framework for understanding evolution.

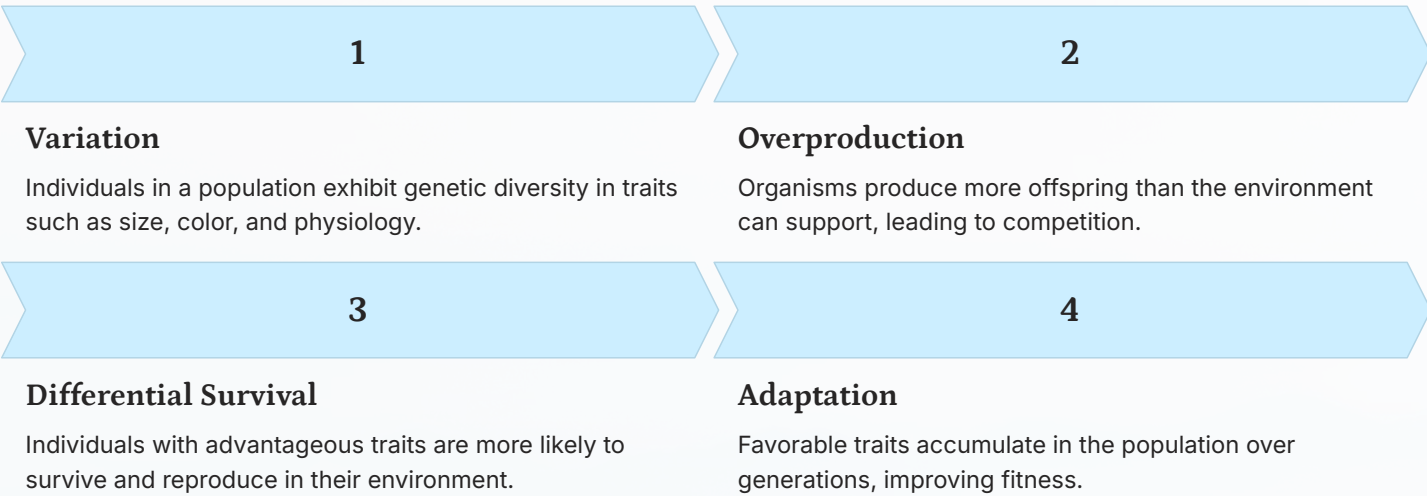
Natural Selection & Speciation

EVOLUTION & PALEOBOTANY

Natural Selection

Natural selection is the primary mechanism of evolution proposed by Charles Darwin. It operates on the principle that individuals within a population exhibit **heritable variation**, and those with traits better suited to their environment are more likely to survive and reproduce, passing those advantageous traits to the next generation. Over time, this differential reproductive success leads to adaptation and, eventually, the emergence of new species.

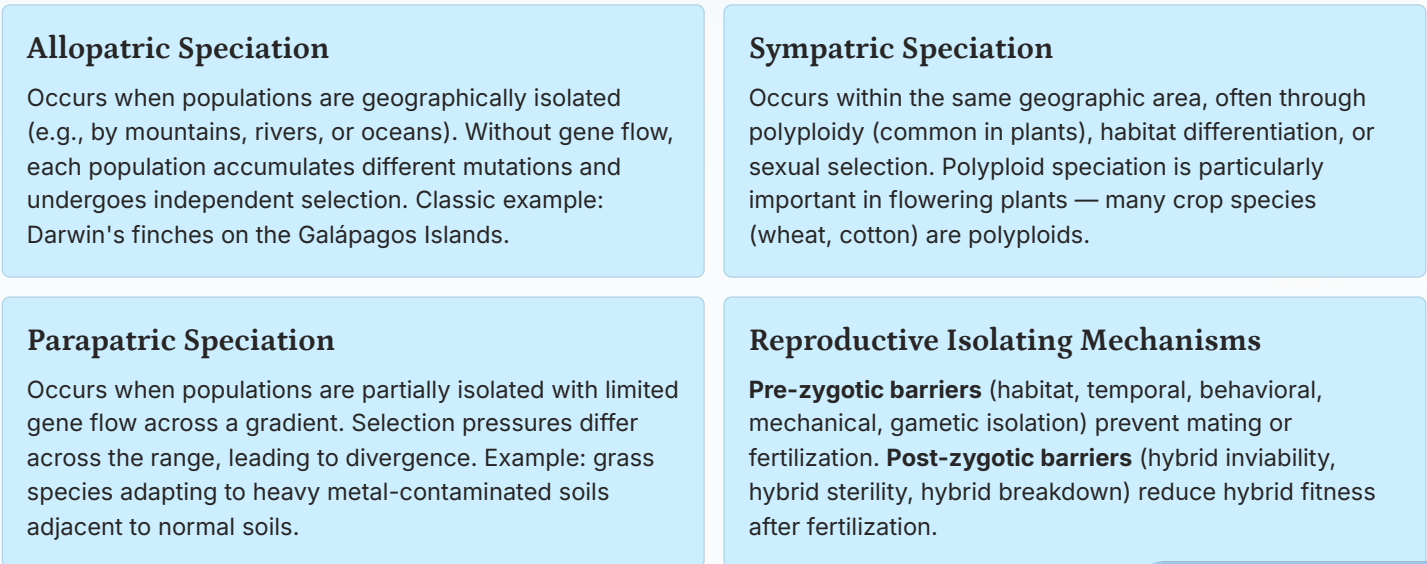
Darwin identified four key observations supporting natural selection: (1) populations produce more offspring than can survive; (2) population sizes remain relatively stable; (3) individuals vary in their traits; and (4) many of these variations are heritable. From these, he concluded that individuals with favorable variations would have a better chance of survival — the famous phrase "**survival of the fittest**" (coined by Herbert Spencer, adopted by Darwin).



Types of natural selection include: **Directional selection** (favors one extreme phenotype, e.g., antibiotic resistance in bacteria), **Stabilizing selection** (favors intermediate phenotypes, e.g., human birth weight), and **Disruptive selection** (favors both extremes, potentially leading to speciation). **Sexual selection**, a special case, involves traits that improve mating success rather than survival, such as the peacock's tail.

Speciation

Speciation is the evolutionary process by which populations evolve to become distinct species. The **Biological Species Concept** (Ernst Mayr, 1942) defines a species as a group of actually or potentially interbreeding natural populations that are reproductively isolated from other such groups. Speciation occurs when gene flow between populations is interrupted, allowing them to diverge genetically.



Fossils & the Geological Time Scale

EVOLUTION & PALEOBOTANY

Fossils: Windows into the Past

A **fossil** is any preserved evidence of past life — including bones, shells, leaves, footprints, and even molecular traces — found in rocks or sediments. Fossils provide the most direct evidence of evolution, revealing how organisms have changed over millions of years. The study of fossils is called **paleontology**, and when focused on plants, **paleobotany**.

Fossilization is a rare event requiring specific conditions: rapid burial, low oxygen, and the presence of hard parts (bones, shells, wood). Common types of fossil preservation include: **petrification** (mineral replacement of organic material), **casts and molds** (impressions left in sediment), **carbonization** (thin carbon films preserving soft tissues), **amber preservation** (organisms trapped in fossilized resin), and **trace fossils** (footprints, burrows, coprolites). **Index fossils** — species that existed for short, well-defined periods — are crucial for dating rock layers and correlating strata across regions.



Relative Dating

Determines the age of fossils by their position in rock layers (strata). The **Law of Superposition** states that in undisturbed sedimentary rock, older layers lie beneath younger ones. Biostratigraphy uses index fossils for correlation.



Radiometric Dating

Measures the decay of radioactive isotopes to determine absolute ages. **Carbon-14 dating** (half-life: 5,730 years) is used for recent fossils (up to ~50,000 years). **Potassium-Argon** and **Uranium-Lead** dating are used for ancient rocks billions of years old.



Molecular Fossils

Biomarkers or chemical fossils are organic compounds (e.g., lipids, pigments) preserved in rocks that indicate the presence of specific organisms. Hopanes indicate cyanobacteria; steranes indicate eukaryotes. These extend the fossil record beyond visible remains.

The Geological Time Scale

The **Geological Time Scale** is a system of chronological dating that relates geological strata to time. It is divided into **eons**, **eras**, **periods**, and **epochs**, based on major changes in Earth's geology, climate, and life forms. The boundaries between units often correspond to mass extinction events or the appearance of new life forms.

Eon	Era	Key Events	Plant Life
Hadean (4.6–4.0 Ga)	—	Earth formation, heavy bombardment	No life
Archean (4.0–2.5 Ga)	—	First prokaryotes, stromatolites	Cyanobacteria; oxygenic photosynthesis begins
Proterozoic (2.5 Ga–541 Ma)	—	Great Oxidation Event; first eukaryotes	Green algae; multicellular organisms appear
—	Ediacaran biota; Snowball Earth episodes	Complex algae diversify	
Phanerozoic (541 Ma–Present)	Paleozoic (541–252 Ma)	Cambrian explosion; colonization of land; Permian extinction	Bryophytes → Pteridophytes → Gymnosperms; first forests (Carboniferous)
Mesozoic (252–66 Ma)	Triassic, Jurassic, Cretaceous; dinosaur dominance; K-Pg extinction	Gymnosperms dominant; angiosperms evolve and radiate in Cretaceous	
Cenozoic (66 Ma–Present)	Tertiary & Quaternary; mammal radiation; ice ages	Angiosperms dominate; grasslands expand; modern flora established	

4.6B

Years Since Earth Formed

The Hadean Eon marks the beginning of Earth's history.

3.5B

Years of Life on Earth

Oldest confirmed microfossils of cyanobacteria.

470M

Years Since Land Plants

First land plants (bryophyte-like) colonized terrestrial habitats.

5

Major Mass Extinctions

Ordovician, Devonian, Permian, Triassic, and Cretaceous events.

Plant Evolution & Fossil Plants

EVOLUTION & PALEOBOTANY

The evolution of plants represents one of the most transformative events in Earth's history. Plants not only changed the atmosphere by producing oxygen but also created the foundation for terrestrial ecosystems, enabling the colonization of land by animals. The plant kingdom evolved from aquatic green algae (Charophytes) through a series of major innovations: the development of **embryos**, **vascular tissue**, **seeds**, and **flowers**.



Green Algae (~1 Billion Years Ago)

Charophyte algae are the closest living relatives of land plants. They share key features: cellulose cell walls, chlorophyll a and b, starch storage, and similar cell division mechanisms (phragmoplast). The transition to land required adaptations to desiccation, UV radiation, and gravity.



Bryophytes (~470 Million Years Ago)

The first land plants were non-vascular, resembling modern mosses, liverworts, and hornworts. They lacked true roots, stems, and leaves but developed a **waxy cuticle** and **stomata** to reduce water loss. They depend on water for reproduction (flagellated sperm). Fossil: *Cooksonia* (Silurian) — one of the earliest known vascular plant precursors.



Pteridophytes (~420 Million Years Ago)

The evolution of **vascular tissue** (xylem and phloem) allowed plants to grow taller and transport water and nutrients efficiently. Ferns, horsetails, and club mosses dominated the Carboniferous period (~360–300 Ma), forming vast swamp forests whose remains became today's coal deposits. Fossil: *Calamites* (giant horsetail), *Lepidodendron* (scale tree).



Gymnosperms (~360 Million Years Ago)

The evolution of the **seed** was a revolutionary adaptation — it protected the embryo, provided nourishment, and allowed dispersal without water. Gymnosperms (naked seeds) include conifers, cycads, ginkgos, and gnetophytes. They dominated the Mesozoic era. Fossil: *Archaeopteris* (first true tree), *Glossopteris* (Gondwana flora, evidence for continental drift).



Angiosperms (~140 Million Years Ago)

Flowering plants evolved in the Early Cretaceous and rapidly diversified, becoming the dominant plant group. Key innovations include **flowers** (attracting pollinators), **fruits** (aiding seed dispersal), and **double fertilization**. Co-evolution with insects drove their radiation. Fossil: *Archaeofructus* (one of the earliest angiosperms, ~125 Ma).

Paleobotanical Evidence

Paleobotany provides critical evidence for plant evolution through the study of fossil plants, pollen, spores, and wood. **Palynology** — the study of pollen and spores — is particularly valuable because pollen grains are highly resistant to decay and can be used to reconstruct ancient vegetation, climate, and even date rock layers. The fossil record reveals major evolutionary transitions: the appearance of vascular tissue, the origin of seeds, and the rise of flowering plants.

Key paleobotanical discoveries include: **Glossopteris flora** — found across South America, Africa, India, Australia, and Antarctica, providing evidence for the supercontinent Gondwana and supporting Wegener's theory of continental drift; **Coal ball petrifications** from the Carboniferous period preserving cellular detail of ancient swamp forests; and **silicified wood** (petrified wood) revealing the anatomy of ancient trees. The **Rhynie Chert** (Scotland, ~400 Ma) is one of the most important fossil sites, preserving early land plants in extraordinary cellular detail, including evidence of early plant-fungal symbiosis (mycorrhizae).

Economic Botany: Cereals, Pulses & Fibers

UMBRELLA 10 — ECONOMIC BOTANY

Economic botany is the study of plants used by humans for food, medicine, fiber, fuel, and other purposes. It bridges botany with agriculture, ecology, and anthropology. The domestication of plants — beginning around 10,000 years ago during the **Neolithic Revolution** — transformed human societies from hunter-gatherers to agricultural civilizations. Today, a small number of plant species provide the majority of the world's caloric intake and industrial raw materials.

Cereals

Cereals are grasses (family **Poaceae**) cultivated for their edible grains (caryopses). They are the primary source of carbohydrates and provide over **50% of the world's food energy**. Major cereals include: **Wheat** (*Triticum aestivum*) — originated in the Fertile Crescent; staple of temperate regions; used for bread, pasta, and biscuits. **Rice** (*Oryza sativa*) — originated in Asia; staple for over half the world's population; Assam is a major rice-growing region with indigenous varieties like *Bao-dhan* (sticky rice) and *Joha* (aromatic rice). **Maize** (*Zea mays*) — originated in Mexico; used for food, fodder, and industrial products. Other important cereals: barley, oats, sorghum, and millets.

Pulses

Pulses are the edible seeds of leguminous plants (family **Fabaceae**). They are rich in protein (20–35%), dietary fiber, and essential minerals, making them crucial for vegetarian diets. Pulses also fix atmospheric nitrogen through symbiosis with *Rhizobium* bacteria, improving soil fertility. Major pulses include: **Lentil** (*Lens culinaris*), **Chickpea** (*Cicer arietinum*), **Pigeon pea** (*Cajanus cajan*), **Green gram** (*Vigna radiata*), **Black gram** (*Vigna mungo*), and **Soybean** (*Glycine max*) — the "miracle bean" with the highest protein content among pulses. India is the world's largest producer and consumer of pulses.

Fibers

Plant fibers are classified as **surface fibers** (cotton — seed hairs of *Gossypium* spp.), **bast fibers** (jute — *Corchorus capsularis* and *C. olitorius*; flax — *Linum usitatissimum*; hemp — *Cannabis sativa*), and **leaf fibers** (sisal — *Agave sisalana*; abaca — *Musa textilis*). **Jute** is particularly important in Assam and West Bengal — it is known as the "golden fiber" and is used for sacks, textiles, and geo-textiles. **Cotton** (*Gossypium hirsutum* and *G. barbadense*) is the world's most important natural fiber crop, with India being one of the largest producers. **Coir** (from coconut husk) is used for ropes, mats, and horticultural purposes.

Cereals

- Wheat — *Triticum aestivum*
- Rice — *Oryza sativa*
- Maize — *Zea mays*
- Barley — *Hordeum vulgare*

Pulses

- Chickpea — *Cicer arietinum*
- Lentil — *Lens culinaris*
- Soybean — *Glycine max*
- Pigeon pea — *Cajanus cajan*

Fibers

- Cotton — *Gossypium* spp.
- Jute — *Corchorus* spp.
- Flax — *Linum usitatissimum*
- Hemp — *Cannabis sativa*

Economic Botany: Oils, Spices & Beverages

UMBRELLA 10 — ECONOMIC BOTANY

Vegetable Oils

Vegetable oils are extracted from seeds, fruits, or nuts and are essential for cooking, nutrition, and industry. They are classified as **non-drying oils** (olive, peanut — used for cooking), **semi-drying oils** (cottonseed, sesame — used in food and industry), and **drying oils** (linseed, tung — used in paints and varnishes). Major oilseed crops include: **Groundnut** (*Arachis hypogaea*) — rich in monounsaturated fats; **Sunflower** (*Helianthus annuus*) — high in vitamin E and linoleic acid; **Mustard** (*Brassica juncea*) — widely used in Assam and eastern India; **Coconut** (*Cocos nucifera*) — saturated fat, used in cooking and cosmetics; and **Oil Palm** (*Elaeis guineensis*) — the highest-yielding oil crop, though environmentally controversial.

Spices

Spices are aromatic plant parts (bark, roots, seeds, fruits, flowers) used to flavor, color, or preserve food. They have been traded for thousands of years and were a major driver of global exploration. India is known as the "Land of Spices," producing over 75 varieties. Key spices include: **Black Pepper** (*Piper nigrum*) — the "King of Spices," native to the Western Ghats; **Cardamom** (*Elettaria cardamomum*) — "Queen of Spices," used in tea and desserts; **Turmeric** (*Curcuma longa*) — contains curcumin, anti-inflammatory properties; **Cinnamon** (*Cinnamomum verum*) — from bark; **Cloves** (*Syzygium aromaticum*) — dried flower buds; **Ginger** (*Zingiber officinale*) — rhizome used in food and medicine; and **Saffron** (*Crocus sativus*) — the world's most expensive spice, from stigmas.

Beverages

Plant-based beverages are among the most economically significant agricultural products globally. **Tea** (*Camellia sinensis*) — originated in China; Assam is the world's largest tea-growing region, producing robust, malty black tea. The tea plant was discovered growing wild in Assam by Robert Bruce in 1823. **Coffee** (*Coffea arabica* and *C. canephora*) — originated in Ethiopia; contains caffeine, a stimulant alkaloid. **Cocoa** (*Theobroma cacao*) — source of chocolate; originated in Central America. **Cola** (*Cola nitida*) — source of caffeine and kola nuts used in beverages. **Maté** (*Ilex paraguariensis*) — a traditional South American stimulant beverage.

Assam's Beverage Heritage

Assam produces over **50% of India's tea** and nearly **17% of the world's tea**. The region's unique climate — high humidity, heavy rainfall, and fertile alluvial soil — produces tea with a distinctive malty flavor and bright color. The **Assam Orthodox** and **CTC** (Crush, Tear, Curl) varieties are globally recognized. Tea cultivation supports millions of workers in the region.

Key Beverage Plants

- **Tea** — *Camellia sinensis* var. *assamica*
- **Coffee** — *Coffea arabica* / *C. canephora*
- **Cocoa** — *Theobroma cacao*
- **Cola** — *Cola nitida*
- **Maté** — *Ilex paraguariensis*

Medicinal Plants & Ethnobotany

UMBRELLA 10 — ECONOMIC BOTANY & ETHNOBOTANY

Medicinal Plants

Medicinal plants have been used for thousands of years across all human cultures. The **World Health Organization** estimates that approximately **80% of the world's population** relies on traditional plant-based medicines for primary healthcare. India's traditional systems — **Ayurveda**, **Unani**, and **Siddha** — document the use of over 7,000 plant species for medicinal purposes. The **Drug Controller General of India** recognizes over 1,200 medicinal plant species.



Tulsi — *Ocimum sanctum*

Known as the "Queen of Herbs" in Ayurveda. Contains eugenol, rosmarinic acid, and ursolic acid. Used for respiratory conditions, immunity enhancement, and stress relief. Considered sacred in Hindu tradition.



Ashwagandha — *Withania somnifera*

An adaptogenic herb used to combat stress, improve vitality, and enhance immunity. Contains withanolides. Used in Ayurvedic formulations for over 3,000 years. Modern research confirms anti-inflammatory and neuroprotective properties.



Neem — *Azadirachta indica*

Known as the "Village Pharmacy." Contains azadirachtin, a potent insecticide. Used for skin diseases, dental care, and as an antimicrobial. Every part of the tree — leaves, bark, seeds, oil — has medicinal or agricultural uses.



Aloe Vera — *Aloe barbadensis*

The gel from leaves is used for wound healing, burns, and skin conditions. Contains polysaccharides, vitamins, and enzymes. Also used in cosmetics and as a laxative (from the latex). One of the most commercially exploited medicinal plants globally.

Ethnobotany

Ethnobotany is the scientific study of the relationships between people and plants — how different cultures use, manage, and perceive plants. The term was coined by John William Harshberger in 1895. Ethnobotany draws from botany, anthropology, ecology, and pharmacology. It documents traditional knowledge about medicinal plants, food plants, ritual plants, and construction materials, often preserving knowledge that is not recorded in scientific literature.

Key applications of ethnobotany include: **Drug discovery** — many modern medicines are derived from plants used in traditional medicine (e.g., aspirin from willow bark, morphine from opium poppy, artemisinin from *Artemisia annua* for malaria); **Conservation** — documenting traditional ecological knowledge helps protect biodiversity and indigenous rights; **Agriculture** — traditional crop varieties and farming practices offer solutions for sustainable agriculture; and **Cultural preservation** — safeguarding indigenous knowledge systems from extinction.

Did You Know? Approximately **25% of modern pharmaceutical drugs** are derived from plants, and over 60% of cancer drugs and 75% of infectious disease treatments have plant-based origins. The Amazon rainforest alone is estimated to contain over 80,000 medicinal plant species, many still unstudied.

Assam's Medicinal Plants & Forest Products

UMBRELLA 10 — ETHNOBOTANY (ASSAM SPECIFIC)

Assam, located in northeastern India, is one of the world's **biodiversity hotspots**. The state encompasses tropical rainforests, grasslands, wetlands, and the Brahmaputra floodplains, supporting extraordinary plant diversity. The traditional communities of Assam — including the Bodos, Karbis, Mizos, Nagas, and others — have developed rich ethnobotanical knowledge over millennia, using hundreds of plant species for medicine, food, construction, and ritual purposes.

Assam Medicinal Plants

The traditional healers of Assam, known as **Bej-Phukriyas** or **Ojhas**, use a vast repertoire of medicinal plants. Many of these are now being scientifically validated. Key medicinal plants of Assam include: **Tejpat** (*Cinnamomum tamala*) — leaf used for digestive ailments and as a spice; **Kalmegh** (*Andrographis paniculata*) — "King of Bitters," used for liver disorders and fever; **Sarpagandha** (*Rauvolfia serpentina*) — source of reserpine, used for hypertension and mental disorders; **Brahmi** (*Bacopa monnieri*) — cognitive enhancer, used in Ayurveda for memory improvement; **Shatavari** (*Asparagus racemosus*) — female reproductive health; and **Khadira** (*Acacia catechu*) — bark used for skin diseases and as an astringent.



Tejpat — *Cinnamomum tamala*

Indigenous to the Himalayan foothills and Assam. Leaves are used in traditional Assamese cuisine and medicine. Essential oil contains eugenol and cinnamaldehyde. Used for digestive problems, respiratory conditions, and as an antimicrobial.



Kalmegh — *Andrographis paniculata*

Widely used in Assamese folk medicine for treating fever, jaundice, and liver disorders. Contains andrographolide, a potent anti-inflammatory compound. The Assamese call it "Kalo Megh" and use it as a bitter tonic.



Sarpagandha — *Rauvolfia serpentina*

Found in the forests of Assam and the Northeast. The root contains reserpine, the first antihypertensive drug isolated from a plant. Also used in traditional medicine for snake bites, mental illness, and insomnia.



Khadira — *Acacia catechu*

The heartwood yields **catechu** (katha), used in betel quid and as a dye. Medicinally used for skin diseases, wounds, and as an astringent. Common in Assam's deciduous forests and traditionally used by tribal communities.

Forest Products of Assam

Assam's forests yield a wide variety of **Non-Timber Forest Products (NTFPs)** that are crucial for the livelihoods of tribal and rural communities. These include: **Medicinal plants** — over 500 species documented; **Edible plants** — ferns, bamboo shoots, mushrooms, and wild vegetables; **Resins and gums** — from *Shorea robusta* (Sal) and *Boswellia* species; **Tendu leaves** — for bidi making; **Honey and beeswax** — from wild bee species; **Lac** — from *Laccifer lacca* insects on host trees; and **Orchids** — Assam has over 300 orchid species, many with medicinal and horticultural value. The **Forest Rights Act (2006)** recognizes the rights of forest-dwelling communities to collect and sell NTFPs.

Bamboo & Tea: Assam's Botanical Icons

UMBRELLA 10 — ASSAM SPECIFIC

Bamboo — The Green Gold of Assam

Bamboo belongs to the family **Poaceae** (subfamily **Bambusoideae**) and comprises over 1,600 species worldwide. Assam is one of India's richest bamboo zones, with over **40 species** recorded. Bamboo is often called "green gold" or "poor man's timber" due to its extraordinary versatility, rapid growth (some species grow up to 1 meter per day), and economic importance to rural and tribal communities.

Key bamboo species of Assam include: **Bambusa tulda** (Bholu Bans) — used for construction and paper industry; **Dendrocalamus hamiltonii** (Baruah Bans) — shoots are edible and highly valued; **Melocanna baccifera** (Muli Bans) — the most abundant species, fruits every 48–50 years in a phenomenon called **gregarious flowering**, followed by mass death of the plants; and **Ochlandra travancorica** — used for reed instruments and paper. Bamboo shoots are a traditional food in Assam, fermented as *Bashor Tenga* or used in curries.

Bamboo Uses in Assam

- Construction — houses, bridges, scaffolding
- Cottage industry — baskets, mats, furniture
- Food — bamboo shoots (fresh and fermented)
- Paper and pulp industry
- Musical instruments (flutes, drums)
- Tribal rituals and crafts
- Erosion control and afforestation

Conservation Concerns

Bamboo forests in Assam face threats from **over-harvesting**, **habitat loss**, and the cyclical devastation caused by gregarious flowering. The **National Bamboo Mission** and various state programs aim to promote sustainable bamboo cultivation and value-added products. Bamboo is now classified as grass (not tree) under Indian forest law, enabling easier cultivation and trade.

Tea — The Lifeline of Assam

Tea (*Camellia sinensis* var. *assamica*) is Assam's most iconic plant and its largest industry. The **Assam tea variety** is distinct from the Chinese variety (*C. sinensis* var. *sinensis*) — it has larger leaves, is more robust, and produces a stronger, maltier brew. Tea was discovered growing wild in Assam by Scottish adventurer **Robert Bruce** in 1823, and the first tea garden, **Chabua**, was established in 1837. Today, Assam has over **800 tea gardens** covering approximately 320,000 hectares, producing over 650 million kg of tea annually.

1823

Year of Discovery

Robert Bruce discovered wild tea plants in Assam.

800+

Tea Gardens

Active tea estates across Assam's tea-growing districts.

17%

World's Tea

Assam's share of global tea production.

2M+

Workers Employed

People directly employed in Assam's tea industry.

The tea industry has profoundly shaped Assam's demography, culture, and economy. Tea garden workers, many descended from tribal communities brought from central India during the British colonial period, form a distinct cultural group. Assam tea holds a **Geographical Indication (GI) tag**, recognizing its unique terroir. The two main types produced are **Orthodox tea** (whole leaf, premium) and **CTC tea** (Crush, Tear, Curl — used in tea bags and blends). The **Assam Tea Festival** and the **Tea Heritage Trail** promote tea tourism in the region.